

Leis de Conservação da Mecânica dos Fluidos aplicadas à Hidrodinâmica do Navio

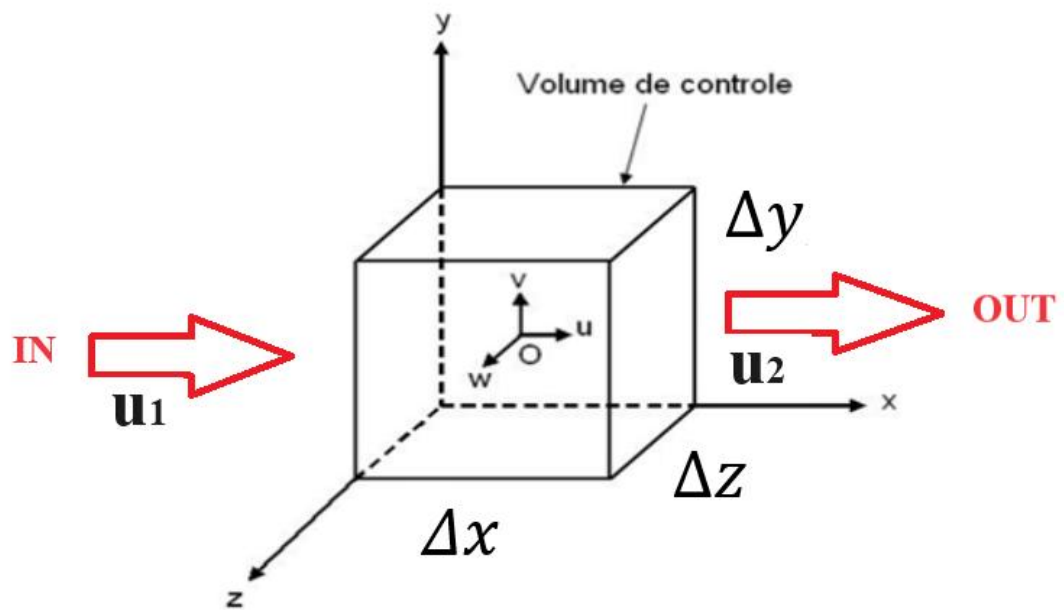
I – Lei da Conservação de Massa

$Vol \quad . \quad .$

$p \quad \tau \quad - \quad -$

$. \quad .$

x 



$$p_{rxw} \quad p_l \quad p_{xer}$$

-

$$p \quad \tau x D$$

$$x \quad \frac{\tau}{w} \quad D \quad C \quad -$$

$$p_{rxw} \quad \tau x D$$

$$p_l \quad \tau x D$$

- -

$$p_{xer} \quad \frac{p}{w} \quad C \frac{\tau}{w} \quad - \quad -$$

$$\frac{\tau x}{}\quad \mathbb{G}\frac{\tau y}{}\quad \frac{\tau z}{}\quad \frac{\tau}{w}$$

$$\tau\qquad \frac{\tau}{w}$$

$$\quad \mathbb{G}\quad $$

$$u,v\,e\,w\qquad\qquad x,y\,e\,z$$

$$\tau\quad -\quad g_D\quad \frac{g}{gw}\quad \tau \mathbb{G} g_{ro}$$

$$\frac{\tau}{w}$$

$$\qquad\qquad\qquad \tau$$

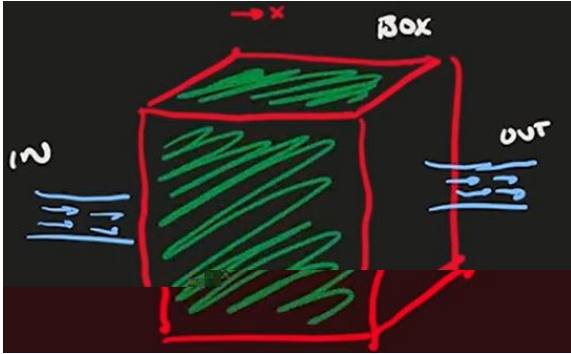
$$\frac{\tau x}{}\quad \mathbb{G}\frac{\tau y}{}\quad \frac{\tau z}{}\quad \tau\quad \frac{x}{}\quad \mathbb{G}\frac{y}{}\quad \frac{z}{}$$

$$\frac{x}{}\quad \mathbb{G}\frac{y}{}\quad \frac{z}{}$$

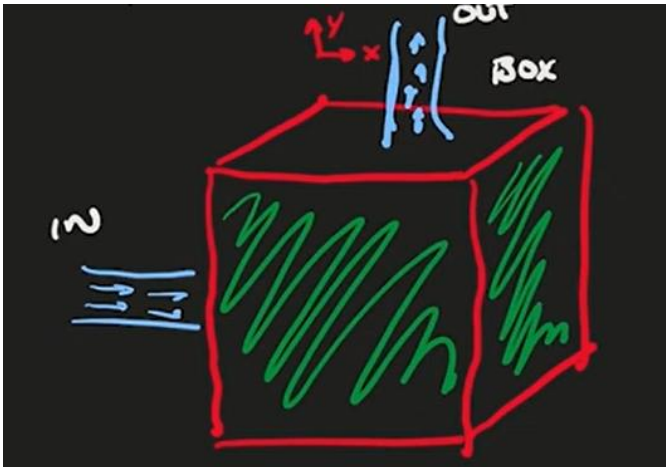
$$-$$

x

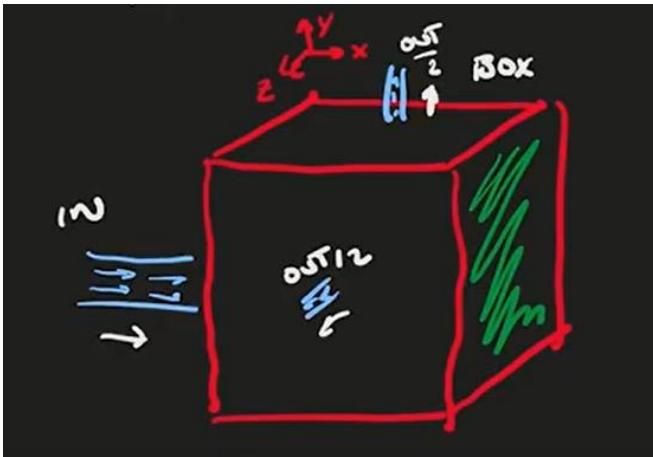
y z



x



G^x G^y



$$\underline{G^x} \quad \underline{G^y} \quad \underline{z}$$